

Miscellaneous problems

20 April 2026 18:13

Generate All Subsets Given a list [1, 2, 3],
print/return all possible subsets (the power set).
Output: [], [1], [2], [3], [1,2], [1,3], [2,3],
[1,2,3]

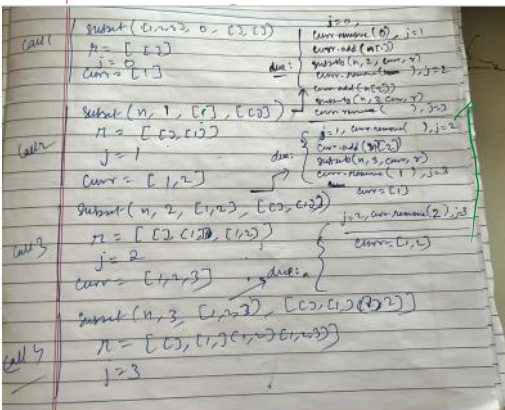
For every element, we have 2 choices.

1. Include it
2. Exclude it.

[] [1] [1,2], [1,2,3] [1,2], [1,3] [2,3]

```
private static void subsets(int[] n, int i,
    List<Integer> current, List<List<Integer>> r) {
    → r.add(new ArrayList<>(current));
    for (int j = i; j < n.length; j++) {
        current.add(n[j]);
        //Recursion
        → subsets(n, j+1, current, r);
        current.remove(current.size() - 1);
    }
}
```

n = [1|2|3]



```
private static void subsets(int[] n, int i,
    List<Integer> current, List<List<Integer>> r) {
    r.add(new ArrayList<>(current));
    for (int j = i; j < n.length; j++) {
        current.add(n[j]);
        //Recursion
        → subsets(n, j+1, current, r);
        → current.remove(current.size() - 1);
    }
}
```

n → [0|1|2]
[1|2|3]

j = 0	j = 1	j = 2	j = 3
current = [1] r = [] i = 0 due: j=0 → current.remove(0)	current = [1, 2] r = [], [1] i = 1 due: j=1 → current.remove(1)	current = [1, 2, 3] r = [], [1], [2] i = 2 due: j=2 → current.remove(2)	current = [1, 2, 3] r = [], [1], [2], [3] i = 3 j < n.length ← false

